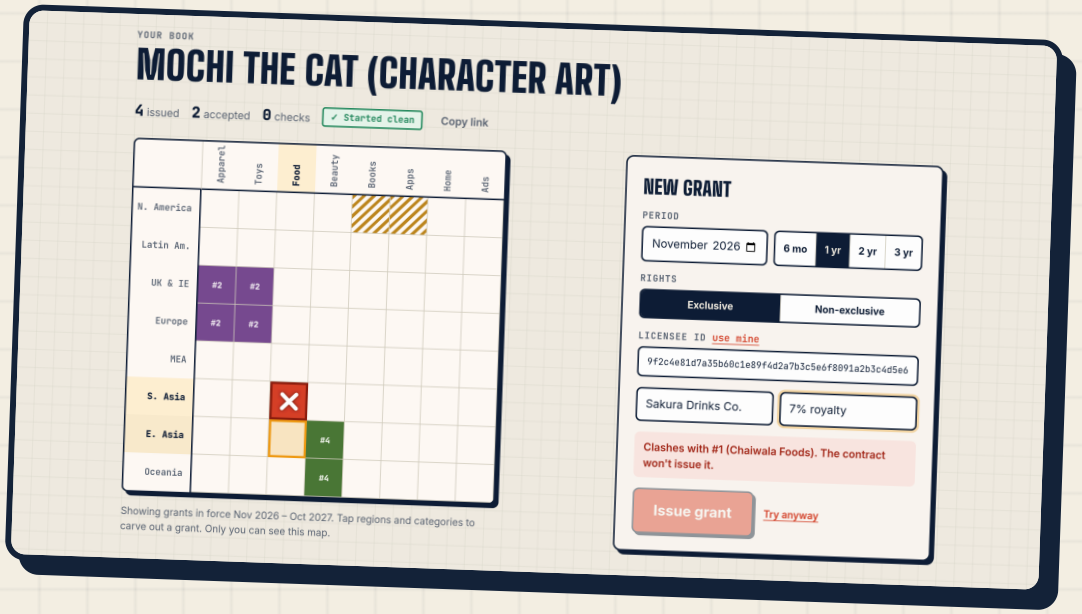
MIDNIGHT BUILDATHON · WAVE 2

CARVE

Sell exclusive rights.
Keep your book private.

Exclusive licensing on Midnight: every grant is proven not to overlap an exclusive one, and nobody sees who bought what.



THE PROBLEM

EXCLUSIVITY IS A PROMISE NOBODY CAN CHECK

$389.8B

Global retail sales of licensed merchandise and services, 2025. Licensing International, Global Licensing Industry Study 2026.

- Almost every deal is carved up by **territory × product category × time**.
- A licensee pays for *"only you sell Mochi drinks in South Asia until 2028"*.
- The licensor's book is confidential: competitors' names, territories and fees. So the licensee takes exclusivity **on trust**.
- Overlapping grants surface after launch, as disputes.

WHY IT ISN'T SOLVED

VERIFIABLE OR PRIVATE. NEVER BOTH.

unverifiable

SPREADSHEETS & CRMS

Private, but nobody outside the agency can check them. Licensees just trust.

leaks the book

A PUBLIC REGISTRY

Anyone can verify, and every competitor gets your client list, territories and fees.

after the fact

LAWYERS & AUDITS

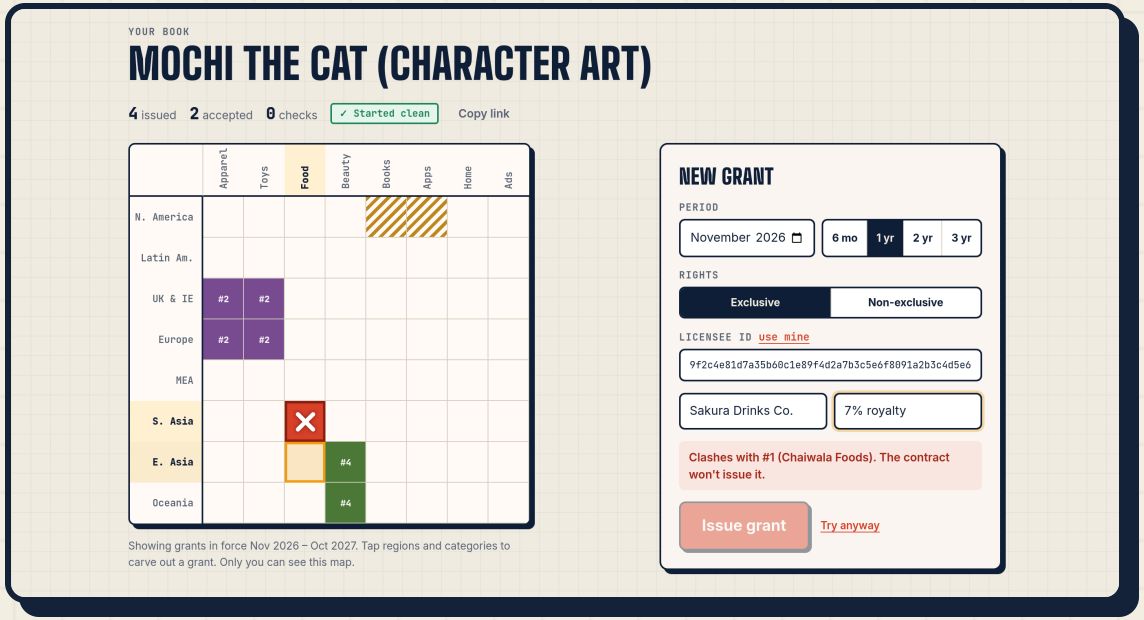
Slow and expensive, and they only start once something has already gone wrong.

Midnight lets the licensor keep the book private and prove things about it.

THE PRODUCT

PROVE THE BOOK. NEVER SHOW IT.

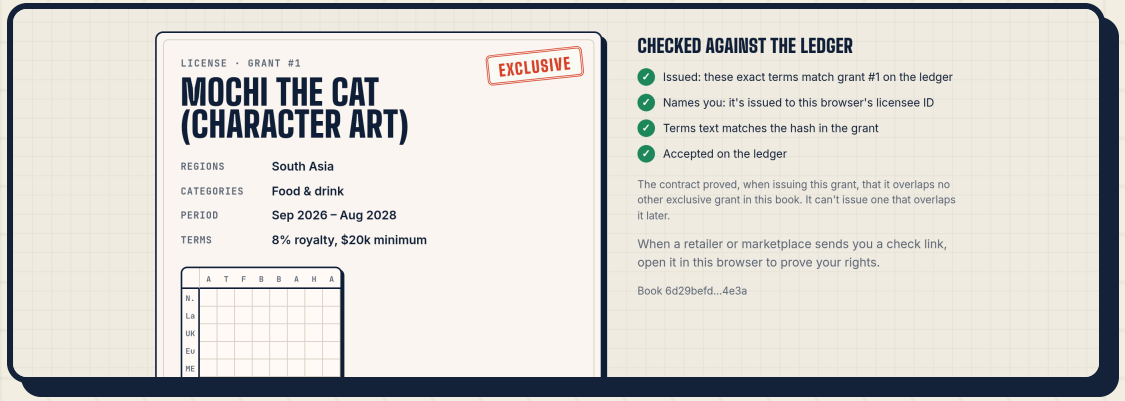
- The licensor's **book** of grants lives on their device. The ledger holds only a commitment.
- Issuing a grant comes with a **zero-knowledge proof** that it overlaps no exclusive grant in the whole book.
- An overlap **can't be issued**: there is no proof for it, even from the licensor's own client.
- The rights map shows the clash before proving. **Try anyway**, and the contract refuses.



THREE PARTIES, THREE PROOFS

EACH SIDE PROVES ONE THING

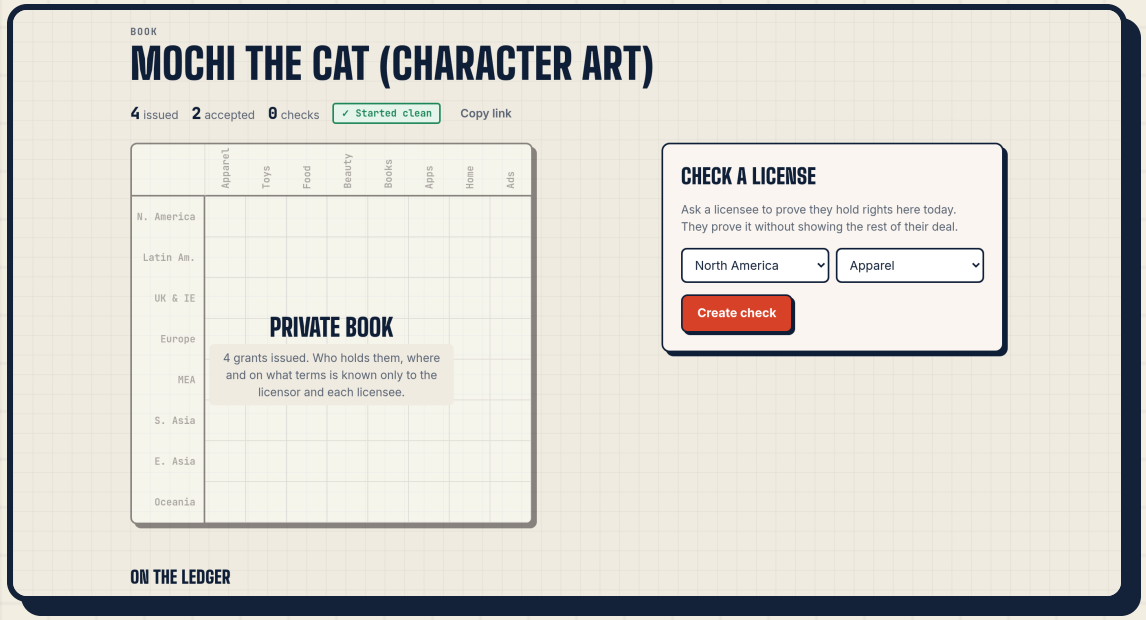
WHO	PROVES IN ZERO KNOWLEDGE	KEEPS PRIVATE
Licensor · `issue`	The grant clashes with no earlier grant, against the book committed on-chain, with nothing removed	Every grant, licensee and term
Licensee · `accept`	"These exact terms were issued, and I'm the one named"	Identity and terms
Licensee · `prove_rights`	"My accepted grant covers *this* region and category *today*"	Other regions, dates, terms, who they are





WHAT THE PUBLIC SEES

COMMITMENTS, COUNTS, ANSWERS



PUBLIC LEDGER	Title, licensor key, book commitment, grant count, one commitment per grant, which are accepted, answers to checks
PRIVATE STATE	Identity secret · the licensor's book (16 grants + salt) · the licensee's grants
OFF-CHAIN	Each grant reaches its licensee as a private link, checked against its commitment

The constructor isn't proven, so the app reads the **deploy transaction's state** and shows `✓ Started clean` only if the book began empty.

BUILT ON MIDNIGHT

ENGINEERING

- **Compact 0.31.1** contract, 3 proven circuits: `issue`, `accept`, `prove_rights`.
- Book commitment is **Poseidon** (`transientHash`): the `issue` key went from **76 MB to 19 MB**. Grant commitments stay SHA-256, domain-separated and salted.
- Time from a witness, bounded by `blockTimeGte` / `blockTimeLt`.
- **28 simulator tests** on the compiled circuits: overlaps, deleted or stale books, impostors, expiry, replayed checks. Mutation-checked.
- React app on **Lace DApp Connector v4**, proving in the wallet. Live at `carve-vert.vercel.app`.
- CLI on midnight-js 4.1.1 for scripted runs. Apache-2.0.

MARKET & ADOPTION

WHO IT'S FOR

FIRST

Independent licensing agencies and character-IP studios with dozens to hundreds of grants per property, and the manufacturers who license from them.

THEN

The same grid shape everywhere: music sync, franchise territories, sports media rights, distribution agreements.

REVENUE

A per-book subscription for licensors, and a per-check fee for marketplaces and retailers that verify listings.

Adoption path: the licensee asks for a Carve link at signing. The agency needs only a browser and a wallet, with no data migration: the book starts with the next deal.

PROGRESS & ROADMAP

WAVE 2 → WAVE 3

BUILT THIS WAVE

- Carve contract, witnesses, simulator, 28 tests
- Rights-map app: issue, accept, check flows on Lace
- Deploy-state verification, license links, check links
- CLI, README, live deploy on Vercel

NEXT

- Merkle-tree books: hundreds of grants per property
- Encrypted book backup, wallet-derived identity
- Property registry bound to the owner's credential
- Sublicensing proven to stay inside the parent grant
- Royalty reports proven against the grant's terms

EXCLUSIVE

CARVE

Exclusive rights you can prove,
from a book nobody can read.

`carve-vert.vercel.app · github.com/argenestel/carve`